

DUT System

Linux + Native Drivers

Logical PCIe Interface

ECAM, BAR, INTx

INTx

DUT Memory

queues, descriptors, payload

FPGA Front End

ECAM Shadow
+ BAR Routes

ECAM / BAR

Transaction Channel
+ Mailbox

ordered events

Interrupt Injector

pending level

DMA Windows

translated addresses

Host Software

Topology + Configuration
Manager

NVMe / NIC
Semantic Adapters

physical control

DMA Address Translator

Completion + Interrupt
Monitor

completion cause

Physical Backend

Selected
Real Device
NVMe SSD or
82580 NIC

device DMA

control / metadata

DMA payload / writebacks